

FUEL STATION SALES AND TANK INVENTORY MANAGEMENT SYSTEM

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PROBLEM STATEMENT

WHY DO WE NEED A DATABASE MANAGEMENT SYSTEM

Fuel stations often rely on manual records for:

- Fuel sales & payments
- Tank inventory & deliveries
- Meter readings & shifts
- Credit customers & invoices

Key Problems

- Difficult stock tracking
- Manual errors
- Difficult sales–inventory reconciliation
- Hard to track credit dues
- Time-consuming reports

A CENTRALIZED RELATIONAL DATABASE TO MANAGE FUEL SALES, INVENTORY, DELIVERIES, SHIFTS, AND CREDIT TRANSACTIONS.

OBJECTIVES

CENTRALIZE FUEL-STATION DATA IN A RELATIONAL DATABASE.

TRACK FUEL INVENTORY ACCURATELY

RECORD SALES AND METER READINGS FOR EACH SHIFT.

MANAGE SUPPLIERS AND FUEL DELIVERIES.

MANAGE CREDIT & PAYMENTS

REDUCE MANUAL ERRORS

GENERATE OPERATIONAL REPORTS

FUNCTIONAL REQUIREMENTS

INVENTORY

- Manage fuel types & tanks
- Track stock
- Record deliveries

SALES

- Manage dispensers
- Capture meter readings
- Record sales
- Manage shifts

CREDIT

- Manage credit customers
- Generate invoices
- Record payments
- Track outstanding balances

REPORTS

- Sales reports
- Inventory reports
- Delivery reports
- Credit reports

SYSTEM USERS

STATION MANAGER

ATTENDANT

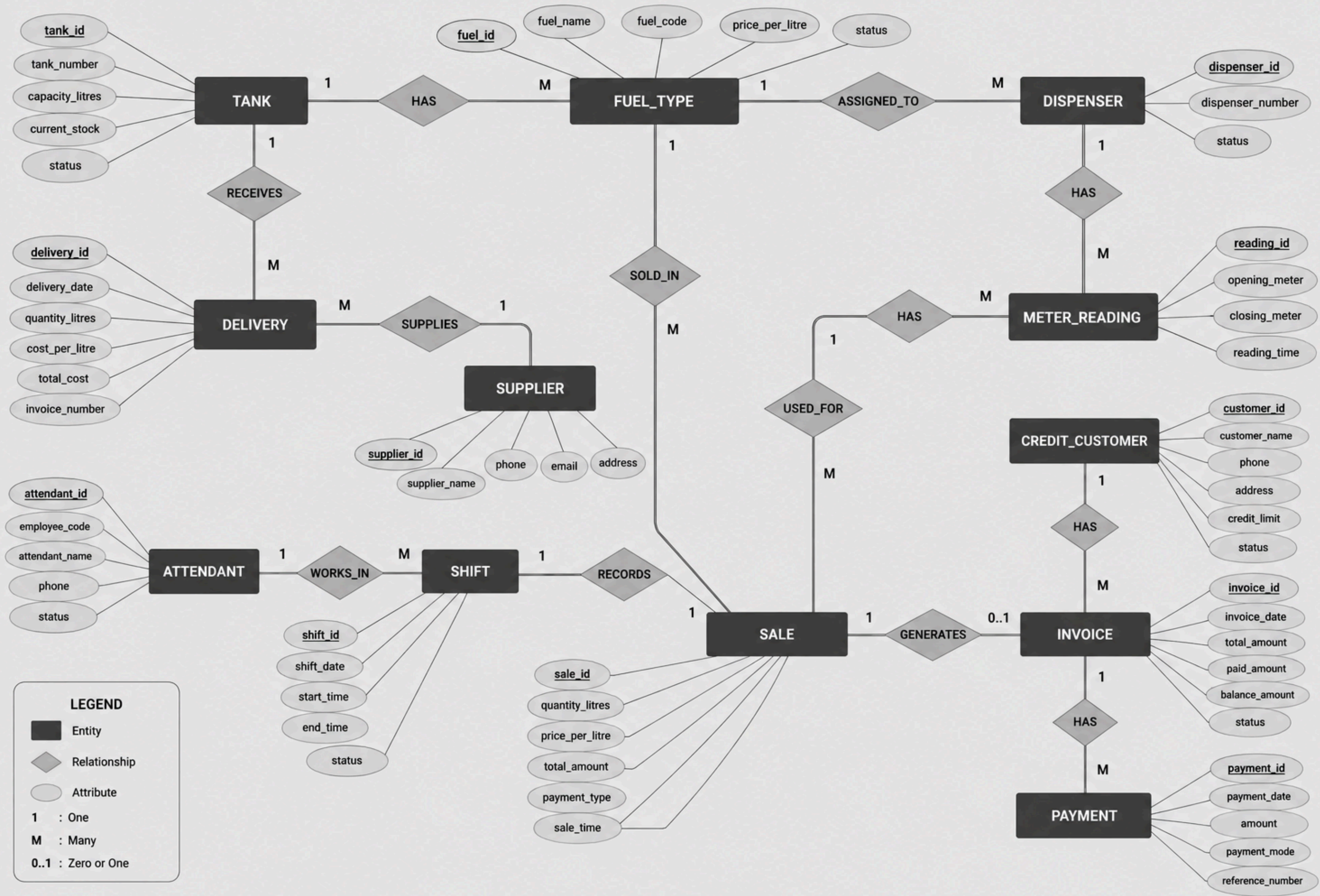
STAFF

ADMINISTRATOR



CORE ENTITIES

1. FUEL_TYPE
2. TANK
3. SUPPLIER
4. DELIVERY
5. DISPENSER
6. METER_READING
7. ATTENDANT
8. SHIFT
9. SALE
10. CREDIT_CUSTOMER
11. INVOICE
12. PAYMENT



ER DIAGRAM — RELATIONSHIPS

#	ENTITY 1	RELATIONSHIP	ENTITY 2	CARDINALITY
1	FUEL_TYPE	HAS	TANK	1 : M
2	FUEL_TYPE	ASSIGNED_TO	DISPENSER	1 : M
3	SUPPLIER	SUPPLIES	DELIVERY	1 : M
4	TANK	RECEIVES	DELIVERY	1 : M
5	ATTENDANT	WORKS_IN	SHIFT	1 : M
6	DISPENSER	HAS	METER_READING	1 : M
7	SHIFT	HAS	METER_READING	1 : M
8	SHIFT	RECORDS	SALE	1 : M
9	DISPENSER	USED_FOR	SALE	1 : M
10	FUEL_TYPE	SOLD_IN	SALE	1 : M
11	CREDIT_CUSTOMER	HAS	INVOICE	1 : M
12	SALE	GENERATES	INVOICE	1 : 0..1
13	INVOICE	HAS	PAYMENT	1 : M

1 = One M = Many 0..1 = Zero or One

INITIAL RELATIONAL SCHEMA

TABLE	PRIMARY KEY	IMPORTANT FOREIGN KEYS
FUEL_TYPE	fuel_id	—
TANK	tank_id	fuel_id
SUPPLIER	supplier_id	—
DELIVERY	delivery_id	supplier_id, tank_id
DISPENSER	dispenser_id	fuel_id
ATTENDANT	attendant_id	—
SHIFT	shift_id	attendant_id
METER_READING	reading_id	dispenser_id, shift_id
SALE	sale_id	shift_id, dispenser_id, fuel_id
CREDIT_CUSTOMER	customer_id	—
INVOICE	invoice_id	customer_id, sale_id
PAYMENT	payment_id	invoice_id

PK = Primary Key FK = Foreign Key